TABLES, LISTINGS AND FIGURES

Table 14.3.16 Exposure-Adjusted Adverse Events — KLV-7702

Kelvaparib (KLV-3308)

Marnholt Oncology GmbH

All displays derive from the analysis datasets locked on 1 May 2019.

Population: safety analysis set (N=58). Content: exposure expressed as person-time, and adverse event counts expressed both as crude incidence and per unit of person-time. This is a supplementary analysis, listed in the statistical analysis plan as output 14.3.16.

Table 14.3.16.1 Extent of Exposure in Patient-Years – Safety Analysis Set

Parameter	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Subjects treated, n	6	9	31	12	58
Total exposure (patient-years)	2.6	3.1	9.7	5.8	21.2
Exposure per subject (patient-years)					
Mean	0.428	0.347	0.313	0.486	0.366
Median	0.528	0.320	0.323	0.452	0.334
Min, max	0.178, 0.646	0.022, 0.843	0.022, 0.652	0.022, 1.227	0.022, 1.227
Duration of treatment (days)					
Total	937	1140	3542	2130	7749
Mean	156.2	126.7	114.3	177.5	133.6

Person-time is derived from the treatment duration recorded for each subject in the subject-level analysis dataset, which agrees with the treatment start and end days in the exposure dataset for every subject in the study. One patient-year is 365.25 days. Duration of treatment in days is the same quantity as summarised in Table 14.1.3, which gives its median, quartiles and range; the totals and means are given here because a rate requires them and Table 14.1.3 does not present them.

Table 14.3.16.2 Adverse Events, Crude Incidence and Exposure-Adjusted Rate, by Dose Cohort – Safety Analysis Set

Parameter	100 mg (N=6)	200 mg (N=9)	300 mg (N=31)	400 mg (N=12)	Total (N=58)
Any adverse event					
Subjects with at least one event, n (%)	6 (100.0)	9 (100.0)	31 (100.0)	12 (100.0)	58 (100.0)
Events, n	35	67	248	85	435
Events per 100 patient-years	1364.3	2146.6	2557.4	1457.6	2050.4
Grade 3 or higher					
Subjects with at least one event, n (%)	3 (50.0)	7 (77.8)	21 (67.7)	10 (83.3)	41 (70.7)
Events, n	4	10	34	12	60
Events per 100 patient-years	155.9	320.4	350.6	205.8	282.8
Serious					
Subjects with at least one event, n (%)	1 (16.7)	2 (22.2)	12 (38.7)	7 (58.3)	22 (37.9)
Events, n	1	3	14	7	25
Events per 100 patient-years	39.0	96.1	144.4	120.0	117.8
Leading to discontinuation of treatment					
Subjects with at least one event, n (%)	1 (16.7)	2 (22.2)	5 (16.1)	2 (16.7)	10 (17.2)
Events, n	1	2	5	2	10
Events per 100 patient-years	39.0	64.1	51.6	34.3	47.1
Fatal					
Subjects with at least one event, n (%)	1 (16.7)	0 (0.0)	4 (12.9)	1 (8.3)	6 (10.3)
Events, n	1	0	4	1	6
Events per 100 patient-years	39.0	0.0	41.2	17.1	28.3
Of special interest					
Subjects with at least one event, n (%)	1 (16.7)	2 (22.2)	10 (32.3)	6 (50.0)	19 (32.8)
Events, n	1	2	11	6	20
Events per 100 patient-years	39.0	64.1	113.4	102.9	94.3

Crude incidence counts each subject once within a category. The exposure-adjusted rate counts every event and divides by the person-time in the column, so the two measures answer different questions and will not agree where follow-up differs between columns. Crude incidence of any adverse event is at or near 100 per cent in every cohort and does not discriminate between them; the corresponding rate does.

Table 14.3.16.3 Exposure and Exposure-Adjusted Rates by Age Category – Safety Analysis Set

Subgroup level	< 65 years (N=21)	>= 65 years (N=37)
Total exposure (patient-years)	6.9	14.3
Exposure per subject (patient-years)	0.331	0.386
Grade 3 or higher, n (%)	14 (66.7)	27 (73.0)
Grade 3 or higher, per 100 PY	244.7	301.3
Serious, n (%)	9 (42.9)	13 (35.1)
Serious, per 100 PY	144.0	105.1
Of special interest, n (%)	4 (19.0)	15 (40.5)
Of special interest, per 100 PY	57.6	112.1

Table 14.3.16.4 Exposure and Exposure-Adjusted Rates by Sex – Safety Analysis Set

Subgroup level	Female (N=21)	Male (N=37)
Total exposure (patient-years)	7.8	13.4
Exposure per subject (patient-years)	0.372	0.362
Grade 3 or higher, n (%)	12 (57.1)	29 (78.4)
Grade 3 or higher, per 100 PY	191.8	335.9
Serious, n (%)	7 (33.3)	15 (40.5)
Serious, per 100 PY	89.5	134.4
Of special interest, n (%)	2 (9.5)	17 (45.9)
Of special interest, per 100 PY	25.6	134.4

PY = patient-years. Each panel covers a subgroup prespecified in the statistical analysis plan whose defining variable is present in the analysis datasets.

Table 14.3.16.5 Exposure and Exposure-Adjusted Rates by ECOG Performance Status – Safety Analysis Set

Subgroup level	0 (N=28)	1 (N=23)	2 (N=7)
Total exposure (patient-years)	9.4	9.7	2.1
Exposure per subject (patient-years)	0.337	0.420	0.301
Grade 3 or higher, n (%)	20 (71.4)	16 (69.6)	5 (71.4)
Grade 3 or higher, per 100 PY	296.4	269.0	285.0
Serious, n (%)	10 (35.7)	9 (39.1)	3 (42.9)
Serious, per 100 PY	116.5	113.8	142.5
Of special interest, n (%)	10 (35.7)	8 (34.8)	1 (14.3)
Of special interest, per 100 PY	116.5	82.8	47.5

Table 14.3.16.6 Exposure and Exposure-Adjusted Rates by Number of Prior Lines of Therapy – Safety Analysis Set

Subgroup level	1 (N=30)	2 (N=19)	3 (N=5)	4 (N=4)
Total exposure (patient-years)	10.7	7.6	1.3	1.5
Exposure per subject (patient-years)	0.358	0.402	0.260	0.383
Grade 3 or higher, n (%)	22 (73.3)	15 (78.9)	3 (60.0)	1 (25.0)
Grade 3 or higher, per 100 PY	297.7	288.1	230.7	196.0
Serious, n (%)	15 (50.0)	6 (31.6)	1 (20.0)	0 (0.0)
Serious, per 100 PY	158.2	91.7	76.9	0.0
Of special interest, n (%)	9 (30.0)	8 (42.1)	1 (20.0)	1 (25.0)
Of special interest, per 100 PY	83.7	117.9	76.9	65.3

PY = patient-years. Each panel covers a subgroup prespecified in the statistical analysis plan whose defining variable is present in the analysis datasets.

Table 14.3.16.7 Exposure and Exposure-Adjusted Rates by BRCA Mutation Status – Safety Analysis Set

Subgroup level	BRCA wild-type (N=31)	BRCA1 mutated (N=15)	BRCA2 mutated (N=12)
Total exposure (patient-years)	10.6	7.0	3.6
Exposure per subject (patient-years)	0.342	0.468	0.299
Grade 3 or higher, n (%)	25 (80.6)	10 (66.7)	6 (50.0)
Grade 3 or higher, per 100 PY	358.3	199.4	222.9
Serious, n (%)	11 (35.5)	6 (40.0)	5 (41.7)
Serious, per 100 PY	122.6	99.7	139.3
Of special interest, n (%)	12 (38.7)	4 (26.7)	3 (25.0)
Of special interest, per 100 PY	122.6	57.0	83.6

Table 14.3.16.8 Exposure and Exposure-Adjusted Rates by Homologous Recombination Status – Safety Analysis Set

Subgroup level	Negative (N=26)	Positive (N=32)
Total exposure (patient-years)	8.8	12.4
Exposure per subject (patient-years)	0.338	0.388
Grade 3 or higher, n (%)	20 (76.9)	21 (65.6)
Grade 3 or higher, per 100 PY	363.8	225.5
Serious, n (%)	9 (34.6)	13 (40.6)
Serious, per 100 PY	125.0	112.7
Of special interest, n (%)	10 (38.5)	9 (28.1)
Of special interest, per 100 PY	125.0	72.5

PY = patient-years. Each panel covers a subgroup prespecified in the statistical analysis plan whose defining variable is present in the analysis datasets.

Notes on the analysis presented

The clinical overview for this programme states that where the duration of exposure differs materially between the groups being compared, crude incidence is not directly comparable and an exposure-adjusted rate is used. No exposure-adjusted rate was presented in the original outputs for this study, and no person-time denominator was published. This output supplies both.

It is a supplementary analysis. The statistical analysis plan (statistical-analysis-plan-klv-7702.pdf) lists it as output 14.3.16, with content "exposure in patient-years and exposure-adjusted event rates, overall and by subgroup"; extent of exposure is output 14.1.3. No hypothesis is tested, no p-value is presented and no multiplicity adjustment applies; every rate here is descriptive.

It is numbered 14.3.16 because the clinical study report's section 14.3 runs to 14.3.15; the number continues that sequence. The crude subgroup tables it should be read against are those at 14.3.10 to 14.3.15.

What the comparison of the two measures shows

Crude incidence of grade 3 or higher events is lowest at 100 mg (50.0 per cent) and highest at 400 mg (83.3 per cent). Exposure-adjusted, the highest rate is at 300 mg (350.6 per 100 patient-years) and 400 mg falls to 205.8, the second lowest of the four. The 400 mg cohort carried 0.486 patient-years per subject against 0.347 at 200 mg, so its subjects had longer in which to accumulate a first event. The crude ordering across dose is not reproduced once exposure is taken into account.

No causal reading follows from either measure. Person-time adjustment removes the effect of differing follow-up on the count of events; it does not adjust for the reasons follow-up differed, and subjects who tolerated treatment stayed on it longer. A cohort with long exposure may therefore have a low adjusted rate partly because its subjects were the ones able to continue. The comparison is between non-randomised groups in an open-label study, and neither column supports a conclusion about the effect of kelvaparib dose on the risk of any event.

Where a stratum is small the rate is unstable, because both the numerator and the person-time in the denominator are small. Rates computed on fewer than ten subjects are presented for completeness and should not be compared with rates from the larger strata.